

Case Study : Bellabeat Fitness Data Analysis

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1. Introduction :

“ This case study analyses the smart device fitness data to unlock new growth opportunities for Bellabeat, a high-tech manufacturer of fitness based products for women. By analyzing how consumers use non-bellabeat smart devices, this report provides high level recommendations for bellabeat’s marketing strategy.

2. Business Task :

The goal of this analysis is to identify the trends in the smart devices usage and understand how these trends inform Bellabeat’s marketing strategy to encourage more activity and better health habits among women.

3. Data Source :

The data used for this analysis is the FitBit Fitness Tracker Data (available on kaggle). It contains personal fitness tracker data from thirty fitbit users, including minute-level output for physical activity, heart rate, and sleep monitoring.

4. Data Processing :

For this project, i choose python because of its high ability to handle large datasets and quality visualizations. My processing steps includes :

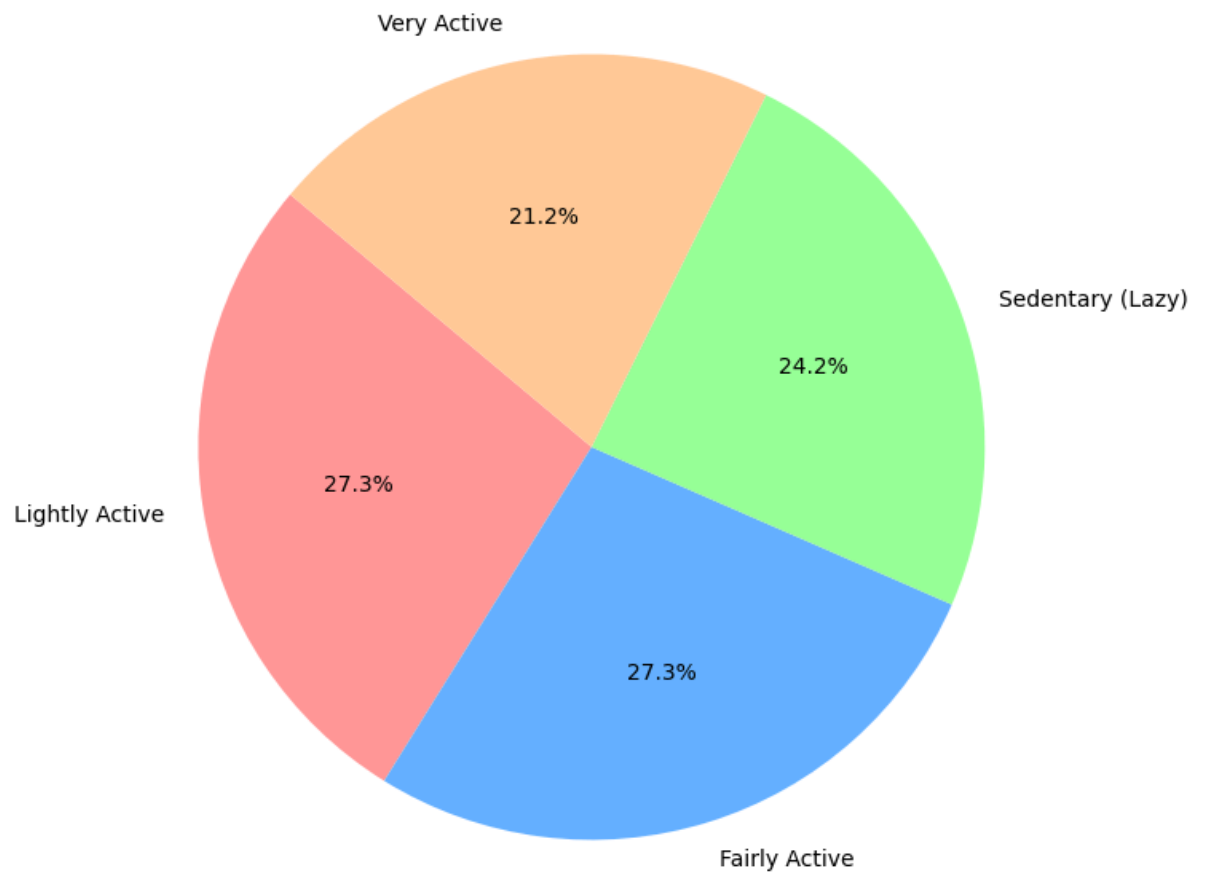
- **Cleaning** : Removes the duplicates and checking for the any missing data.
- **Formatting** : Converting the the date column into the correct format for analysis.
- **Merging** : Combining the daily activity data with the sleep data to find the correlation between the movement and rest.

5. Key Insights & Visualizations

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A. Distribution of User Activity Level :

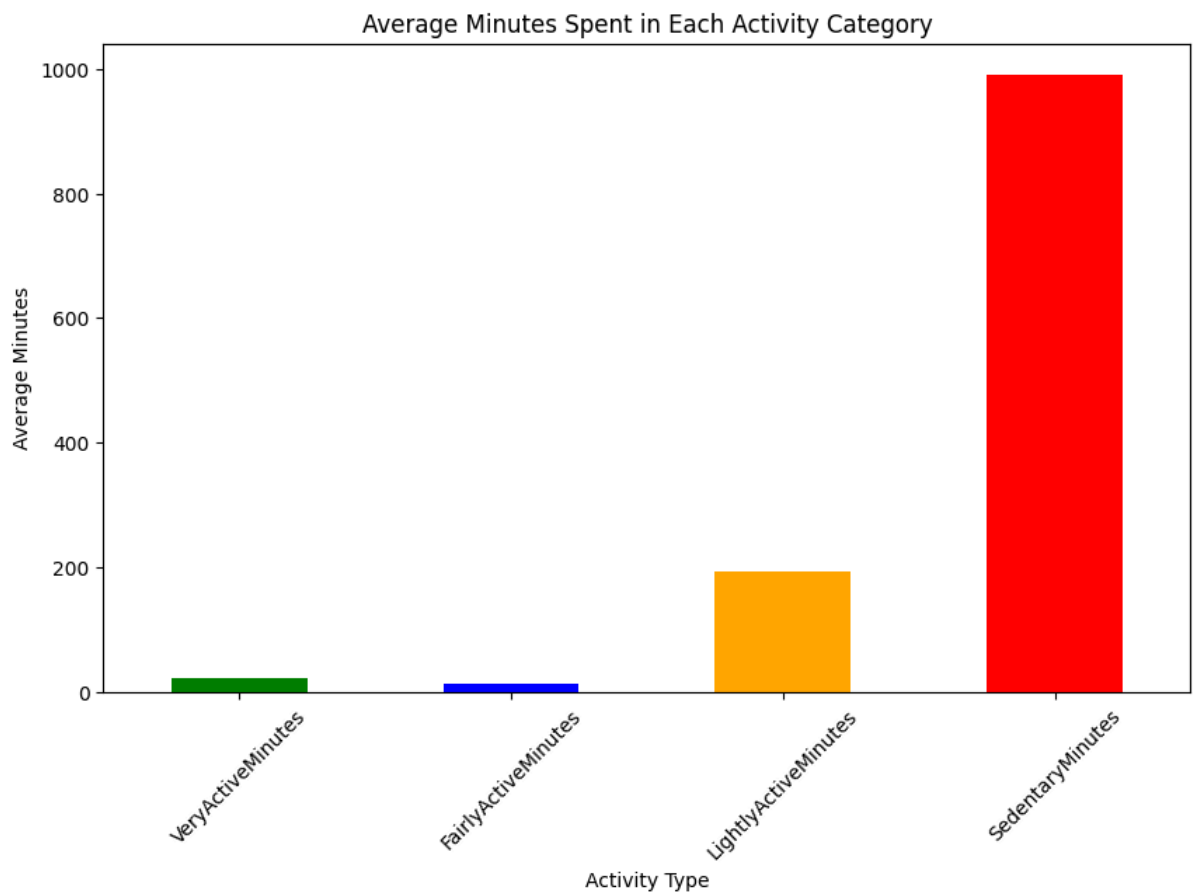
Distribution of Bellabeat User Activity Levels



Obseravtion : The analysis clearly shows that large number of users falls under 'Lightly Active' and 'Sedentary' categories. Only a small percentage of user hitting the target of 10,000 steps per day.

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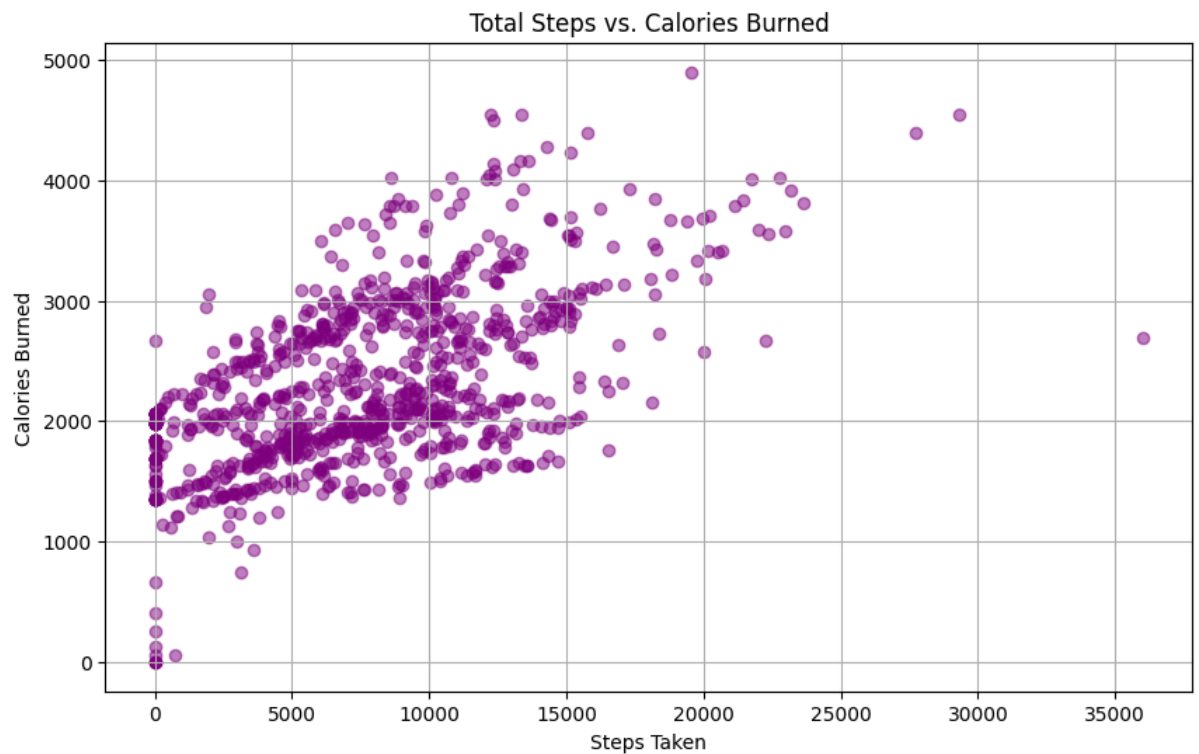
B. Daily Average Activity Minutes :



Observation : The data shows a massive gap between activity types. Users spend an average of 991 minutes sedentary (lazy) compared to 21 minutes of very active users. This highlights a huge opportunity for Bellabeat to encourage the users to break up long periods of sitting.

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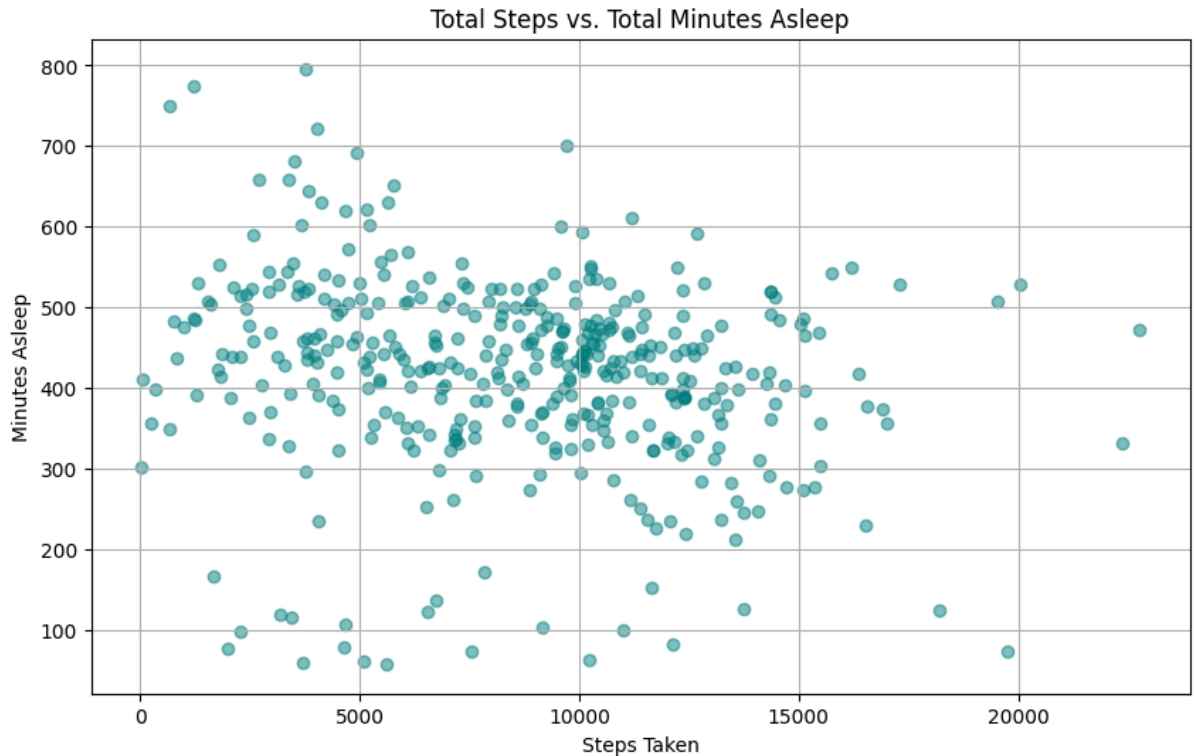
C. Total Steps Vs. Calories Burned :



Observation : There is a clear positive correlation between total steps and calories burned. This confirmed that increasing step counts is an effective strategy for users looking to manage their weight and energy burn.

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D. Relationship Between Activity and Sleep :



Observation : Interestingly the data shows no strong correlation between the number of steps taken and total minutes asleep. This means walking more doesn't automatically mean a user will sleep more.

6. Final Recommendations & Conclusion :

- **Nudge the Sedentary :** Implement "Inactivity Alerts" to encourage users to move during the workday.
- **Gamify the Journey :** Create "Daily Goal" badges for the lightly active group to help them reach 10,000 steps.
- **Beyond Fitness :** Expand the app's sleep features to include guided relaxation, since activity levels alone don't dictate sleep duration.

Conclusion : By leveraging these data driven insights Bellabeat can transition from a simple tracking tool to a personalized wellness coach, creating stronger loyalty among its users.